Cape Peninsula University of Technology

FACULTY OF INFORMATICS & DESIGN

ADVANCED DIPLOMA in INFORMATION & COMMUNICATIONS TECHNOLOGY

Subject Guide

APPLICATIONS DEVELOPMENT THEORY 4

SUBJECT GUIDE FOR:	2025
COURSE CODE:	APT470S
NQF LEVEL:	7
PRE-REQUISITE SUBJECTS:	Applications Development Theory 3 or equivalent

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1. INTRODUCTION

Welcome to Applications Development Theory 4. This subject follows on from the diploma subject of the same name where the foundation for software development was set. Applications Development Theory 4 focuses on the following core topics related to the design, and development of quality software artifacts namely Software Requirements Engineering, Software Design based on the ISO/IEC 12207-2017, and ISO/IEC 15288-2015 standard; Software Configuration Management and Software Quality.

This Study Guide provides an overview of the course, its objectives, expected learning outcomes, and the mode of assessment that will be engaged in the administration of this course. The relevant resources and important dates are also listed.

2. GENERAL INFORMATION

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Lecturer	Prof A Ajibesin	Engineering		ajibesina@cput.ac.za
Domain Coordinator	T Makhurane	Engineering 2.11	(021) 460-3348	makhuranet@cput.ac.za
Head of Department	Dr. T Ncubukezi	Engineering 2.30	(021) 460-3713	NcubukeziT@cput.ac.za
Secretary	N. Allie	Engineering 2.1	(021) 460-3010	allien@cput.ac.za

2.2 Class Times and Venues

As published on http://myclassroom.cput.ac.za and notice boards. This subject consists of two periods per week.

2.3 Learner Management System (LMS)

Blackboard is used for general communications and announcements, posting of course material, assignments, and announcements. It is the responsibility of the learner to check Blackboard regularly at https://myclassroom.cput.ac.za to remain up to date with all subject matter. Notifications will also be sent from Blackboard to the university's official email accounts for students at their ***mycput* email addresses**.

2.4 Queries (CRIMS System)

CRIMS system is an online platform to help students open tickets and raise subject and other academic-related queries. Students must not open a CRIMS ticket if the matter can be addressed with the Lecturer/Subject Coordinator directly. You can access this system at https://enquiries.cput.ac.za

2.5 Recognition of Prior Learning (RPL)

In a situation where there is a claim of previous learning that equates to a qualification in Information Technology, the recognition of such qualification will be at the discretion of the higher education institution. A structured process for the assessment of individual learners will be followed, vis-a-vis the exit level outcomes of the qualification on a case-by-case basis. Independent assessors will moderate this process and the assessment of individual cases.

In this course, we recognize RPL by granting credit to learners who demonstrate both theoretical and practical knowledge of software engineering and develop large software-intensive systems. This is not to be confused with an exemption granted for a similar subject studied at another recognized tertiary institution.

3 SUBJECT INFORMATION

3.1 Subject Credits

This subject has 20 credits at level 7 on the NQF. This means that the average student will take 200 hours per year to master the outcomes of the subject. Of this 30% of the time will be contact hours and the remainder will be self-study and other tasks, assignments, and tests. For further information on this please refer to the Faculty Prospectus at http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus

3.2 Length of the course

METHOD	NOTIONAL HOURS (1 Notional hour = 10 hours)
Formal Lectures	6
Practical / Assignments	4
Self-Study	8
Other	2
TOTAL	**20**

3.3 Learning Outcomes

The learning outcomes associated with this subject include the ability to –

- Recognise, articulate, and compare various models in advanced software development
- Understand and make comparisons among advanced techniques in the design of software systems
- Identify and compare tools required in effective software development and maintenance
- Effectively communicate and evaluate their work and that of their peers

3.4 Lecture Delivery Methods

The mode of lecture delivery will be based on the following: online lectures, mandatory and other readings, assignments (Group and individual).

3.5 Prescribed Textbooks

Software Engineering, by Ian Sommerville 10th edition, Pearson Education Limited, 2016.

and

Software Engineering – A Practitioner's Approach 8th edition Roger Pressman McGraw Hill Education (2015).

3.6. Library and Readings

Additional reading is an essential part of the course, particularly for topics not covered in the prescribed textbooks. Students are expected to make use of the library and the Internet as additional sources of informaiton.

4. ASSESSMENT

4.1 Assessment policy and regulations

This subject has four assessments. Assessments are a combination of Blackboard online tests, Blackboard quizzes and assignments. Assignments are administered via Blackboard.

Test queries: The last day for any queries regarding a test is one week after the test was discussed in class. Thereafter the test mark cannot be changed.

Absence from a test: If you are absent from a test, a valid medical certificate must be submitted to the faculty Customer Relationship Information Management system (CRIMs) at https://enquiries.cput.ac.za/. This should be done ASAP and preferably within one week after the test was written. You must also inform your lecturer on the day of the test that you are ill.

Due dates must be strictly adhered to. No late submissions are allowed for any subject in this Advanced Diploma.

4.2 Assessment Opportunities:

There will be one test per term. An Exception/Sick test will be carried out at the end of the semester. Students may qualify for one Exception/Sick test per subject per year.

4.3 Assessment Breakdown and Weights

ASSESSMENT BREAKDOWN AND WEIGHTS							
ASSESSMENT CRITERIA	1st TERM	2nd TERM		3rd TERM	4th TERM		TOTALS
Nature of assessment	**Class Tests and Assignments**		**Written Exam**	**Class Tests**		**Written Exam**	
Will assessment be included in this term?	Yes	Yes	Yes	Yes	Yes	Yes	**100%**
Total assessment weight for the term	10%	10%	**30%**	10%	10%	**30%**	
Will the assessment be moderated?	No	No	**Yes**	No	No	**Yes**	

4.4 Tests and Assignment Guidelines

All tests are individual assessments. Assignments may either be done individually or as part of a group. A maximum of 4 students are allowed per group. All work must be carried out in a way that ensures a contribution from each member. The assignments are detailed in the Assignment Guide on Blackboard. Deadlines for submission must be adhered to. No late submissions will be accepted.

5 TOPIC ORGANIZATION and SCHEDULE

No	Topic	Week of
Term 1	THEME 1: THE SOFTWARE PROCESS and REQUIREMENTS ENGINEERING	
1	The nature of software and Introduction to Software Engineering	3/2/2025
2	Software Processes and Models	10/2/2025
3	Agile Development	17/2/2025
4	Human aspects of Software Engineering	24/2/2025
5	Requirements Engineering	3/3/2025
	Class Tests and Assignments for Term 1	
Term 2	THEME 2: MODELLING - SYSTEMS DESIGN and DEVELOPMENT	
6	Requirements Modelling	24/3/2025
7	Design Concepts	7/4/2025
8	Architectural Design	21/4/2025
9	Systems Design – component, UI, webApp design, mobileApp design,	5/5/2025
10	Reuse	19/5/2025
	Assignments, Class Tests and Exam on Semester 1 work	
Term 3	THEME 3: QUALITY MANAGEMENT	
11	Quality Concepts,	14/7/2025
12	Review Techniques	21/7/2025
13	Software Quality Assurance, Software Testing Strategies	4/8/2025
14	Software Testing – traditional, OO, webApp, mobileApp testing	11/8/2025
	Class Tests on Term 3 work	
Term 4	THEME 4: SECURITY ENGINEERING, SOFTWARE DEPENDABILITY, PROJECT MANAGEMENT and TRENDS	
15	Security Engineering	15/9/2025
16	Software Configuration Management	22/9/2025
17	Project – management, planning, configuration, risk management and process improvement	6/10/2025
18	Emerging Trends in Software Engineering	20/10/2025
	Class Tests on Term 4 work and Exam on Semester 2 work	

6. STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY
STUDENT ACKNOWLEDGEMENT
2025
ADVANCED DIPLOMA: INFORMATION and COMMUNICATIONS TECHNOLOGY
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ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **APPLICATIONS DEVELOPMENT THEORY 4** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname: ………………………………………………………… (Print)

Student Number: …………………………………

Contact number: ……………………… (home/res) Contact number: ……………………… (cell)

Signature: ……………………… Date: ………………………

